

RECOVERY OF CHROMIUM FROM SPENT CHROME LIQUOR

1. Objective

To recover chromium from spent (exhausted) chrome tanning liquor by precipitating it as chromium hydroxide, $\text{Cr}(\text{OH})_3$, through controlled dosing of 0.5 M sodium hydroxide, and to determine the approximate volume of alkali required to reach the optimum precipitation pH.

2. Principle

Spent chrome liquor from the tanyard is the exhausted chrome-tanning float left after hides have taken up most of the trivalent chromium (Cr^{3+}) they can absorb. It is strongly acidic ($\text{pH} \approx 3\text{--}4$) and still carries a substantial residual load of dissolved Cr^{3+} — typically several thousand mg/L — as basic chromium sulphate. Left untreated, this is one of the most toxic and regulated components of tannery effluent.

Dissolved Cr^{3+} can be converted to an insoluble, settleable solid by raising the pH with an alkali. Trivalent chromium hydroxide is amphoteric: it is highly soluble at low pH, reaches a solubility minimum around pH 8–9, and starts to redissolve as soluble chromite (CrO_2^-) if the pH is pushed much above ≈ 10 . The precipitation reaction is:



Because the target window (pH 8–8.5) is narrow, the alkali must be added gradually with continuous monitoring — dosing too little leaves chromium in solution, and dosing too much redissolves it. The recovered $\text{Cr}(\text{OH})_3$ sludge is not simply a waste: it can be redissolved in dilute sulphuric acid to regenerate a chromium sulphate solution, which is the basis of chrome recovery and reuse programmes in tanneries (a cleaner-production practice, since it cuts both raw chemical purchase and effluent chromium load).

3. Materials and Apparatus

Item	Specification / Quantity
Spent chrome tanning liquor	400 mL (as received, or synthetic $\approx 2,000\text{--}4,000$ mg/L Cr^{3+} , pH 3–4)
Sodium hydroxide solution	0.5 M NaOH, $\approx 250\text{--}300$ mL (see §4 for preparation)
Beaker	500 mL or 1000 mL, borosilicate glass
Stirrer	Magnetic stirrer with stir bar, or a glass stirring rod
pH meter	Calibrated bench or portable pH meter (or universal pH paper as a budget alternative)
Burette / graduated dropper	50 mL burette with stand and clamp, for controlled dosing of NaOH
Filter paper and funnel	Whatman No. 1 (or equivalent), with filter stand / conical flask

Item	Specification / Quantity
Wash bottle	Filled with distilled water
Weighing balance	Optional — for weighing recovered dry sludge
Oven	Optional — 60–80°C, for drying the recovered precipitate
Personal protective equipment	Nitrile gloves, safety goggles, lab coat

4. Preparation of 0.5 M NaOH

Molar mass of NaOH $\approx$ 40 g/mol. For 500 mL of 0.5 M solution:

$$\text{Mass} = \text{Molarity} \times \text{Volume(L)} \times \text{Molar mass} = 0.5 \times 0.5 \times 40 = 10.0 \text{ g}$$

Weigh 10.0 g of NaOH pellets and dissolve in about 400 mL of distilled water in a beaker, stirring continuously.

Dissolution is strongly exothermic — add the pellets slowly and allow the solution to cool before making up to the final 500 mL mark in a volumetric flask. Label the flask clearly as “0.5 M NaOH”.

5. Safety Precautions

- **NaOH is corrosive:** wear gloves and goggles at all times when preparing or dosing it; avoid skin and eye contact and never add water to solid NaOH in a closed vessel.
- **Chromium exposure:** spent chrome liquor contains trivalent chromium, Cr(III), which is far less hazardous than hexavalent Cr(VI) but is still a skin and eye irritant — avoid direct contact and wash off any splashes immediately.
- **Ventilation:** carry out the experiment in a well-ventilated lab or fume hood.
- **Waste disposal:** the recovered Cr(OH)₃ sludge and any leftover liquor are hazardous waste — collect them in a labelled container for controlled disposal or reuse; never pour down the drain.
- **General:** wear a lab coat, tie back loose hair/clothing, and wash hands thoroughly at the end of the experiment.

6. Procedure

1. Measure 400 mL of spent chrome liquor using a measuring cylinder and transfer it into a 500–1000 mL beaker.
2. Record the initial pH and appearance of the liquor (typically a pale bluish-green solution, $\text{pH} \approx 3\text{--}4$).
3. Place the beaker on a magnetic stirrer (or stir manually with a glass rod) and begin gentle, continuous stirring.
4. Fill the burette with the 0.5 M NaOH solution and note the starting (zero) reading.
5. Add the NaOH slowly, drop by drop, into the stirred liquor, monitoring the pH continuously.
6. Record the burette volume at pH 4, 5, 6, 7, 8, and 8.5. Note the pH at which a visible precipitate first appears (usually around pH 5–6, seen as a pale bluish-grey cloudiness).
7. Slow the addition further as the pH approaches 8, adding NaOH dropwise and allowing the reading to stabilise between additions. **Stop dosing once the pH holds steady at 8.0–8.5** — the optimum window for minimum $\text{Cr}(\text{OH})_3$ solubility. Do not let the pH exceed ≈ 10 , or the chromium will begin to redissolve as chromite.
8. Continue gentle stirring for a further 5–10 minutes at the target pH to allow the floc to fully develop.
9. Stop stirring and let the beaker stand undisturbed for 20–30 minutes so the precipitate can settle, leaving a clearer supernatant above the settled sludge.
10. Filter the contents through filter paper, collecting the clear/pale filtrate (treated water) separately from the retained grey-green $\text{Cr}(\text{OH})_3$ sludge on the paper.
11. Wash the sludge on the filter with a small amount of distilled water from the wash bottle to rinse out residual dissolved salts (mainly sodium sulphate).
12. Optional — dry and weigh: dry the recovered sludge in an oven at $60\text{--}80^\circ\text{C}$ until a constant weight is reached, then weigh it to determine the mass of $\text{Cr}(\text{OH})_3$ recovered.
13. Optional — demonstrate reuse: redissolve a small portion of the moist sludge in dilute ($\approx 10\%$) sulphuric acid while stirring. It should turn back into a clear bluish-green chromium sulphate solution, demonstrating that the recovered chromium can be regenerated as fresh tanning liquor.

7. Observation Table

Copy this table into your notebook and complete it as you carry out §6, steps 4–8.

Volume of 0.5 M NaOH added (mL)	pH	Observation (colour / turbidity / precipitate)
0 (initial)		
	4	
	5	
	6	
	7	

Volume of 0.5 M NaOH added (mL)	pH	Observation (colour / turbidity / precipitate)
	8	
	8.5 (final / target)	

8. Calculation

The theoretical NaOH requirement can be estimated from the chromium content of the liquor, if known or measured. This gives a useful benchmark to compare against the volume actually consumed in §6 — the two will not match exactly, because the liquor also contains free residual sulphuric acid and other buffering species that consume alkali before precipitation even begins. The actual dose is always set experimentally, by titrating to the target pH as you did in the procedure.

Worked example — theoretical NaOH requirement

Assume the 400 mL spent liquor contains 3,000 mg/L Cr^{3+} (a typical value for exhausted chrome-tan float; measure or confirm this for your actual sample).

Mass of Cr^{3+} in 400 mL = $3,000 \text{ mg/L} \times 0.400 \text{ L} = \mathbf{1,200 \text{ mg} = 1.2 \text{ g}}$

Moles of $\text{Cr}^{3+} = 1.2 \text{ g} \div 52 \text{ g/mol}$ (atomic mass of Cr) = **0.0231 mol**

From $\text{Cr}^{3+} + 3\text{OH}^- \rightarrow \text{Cr}(\text{OH})_3$, moles of OH^- needed = $3 \times 0.0231 = \mathbf{0.0692 \text{ mol}}$

Volume of 0.5 M NaOH required = $0.0692 \text{ mol} \div 0.5 \text{ mol/L} \approx \mathbf{138 \text{ mL (theoretical, chromium precipitation only)}}$

In practice, expect to use noticeably more than this — the difference is the alkali consumed neutralising residual free acid in the liquor before the pH begins to rise toward the precipitation window.

If you have a way to measure Cr^{3+} concentration in your own sample (e.g. titration or spectrophotometric analysis before the experiment), repeat this calculation with your measured value instead of the assumed 3,000 mg/L.

9. Results and Discussion

Using your observation table and the volumes recorded, report and discuss:

- The pH at which precipitation was first visually observed, and the pH at which it was judged complete (target 8.0–8.5).
- The total volume of 0.5 M NaOH consumed to reach the target pH, and the corresponding NaOH dose per litre of liquor treated.
- How the actual NaOH consumption compares with the theoretical value calculated in §8, and why they differ.
- If weighed: the mass of dried $\text{Cr}(\text{OH})_3$ sludge recovered, and (if the initial Cr content is known) the approximate percentage recovery.
- The appearance of the filtrate versus the original liquor — has the strong colour and turbidity been substantially removed?

- If the reuse step (§6, step 13) was performed: whether the sludge redissolved cleanly in dilute sulphuric acid, and what this demonstrates about chrome recovery as a cleaner-production practice.

10. Precautions and Tips for a Clean Result

- Add NaOH slowly and in small increments as the pH approaches 7–8 — the pH rises very sharply near the end-point, and it is easy to overshoot past 8.5 if dosing too fast.
- Keep stirring continuous throughout dosing so the alkali is evenly distributed and no local over-dosing (and re-dissolution) occurs at the point of addition.
- Allow the pH reading to stabilise for a few seconds after each addition before recording it or adding more.
- Do not exceed pH ≈ 10 : chromium hydroxide is amphoteric and will start to redissolve as soluble chromite at high pH, lowering your recovery.
- Give the precipitate adequate settling time (≥ 20 minutes) before filtering — filtering too soon will clog the paper and slow filtration.
- If using pH paper instead of a meter, take several strips per reading and cross-check colour changes carefully near the pH 7–9 range, where colour bands are close together.

References

1. Standard tannery-effluent chromium precipitation practice, following the reaction $\text{Cr}^{3+} + 3\text{OH}^- \rightarrow \text{Cr}(\text{OH})_3\downarrow$, with a solubility minimum around pH 8–9 (see standard wastewater engineering references, e.g. Tchobanoglous, Burton & Stensel, *Wastewater Engineering: Treatment and Resource Recovery*, McGraw-Hill).
2. Chrome recovery and reuse as a cleaner-production practice in leather processing: Scholz, W., *Key Technologies for Tannery Wastewater Treatment* (IULTCS); UNIDO, *Introduction to Treatment of Tannery Effluents* / Cleaner Production guidance.

This is a simple, single-session bench experiment suitable for an undergraduate lab. Actual tannery-scale chrome-recovery plants follow the same chemistry but add mechanical mixing, sludge dewatering (filter press) and acid-redissolution stages at larger scale.